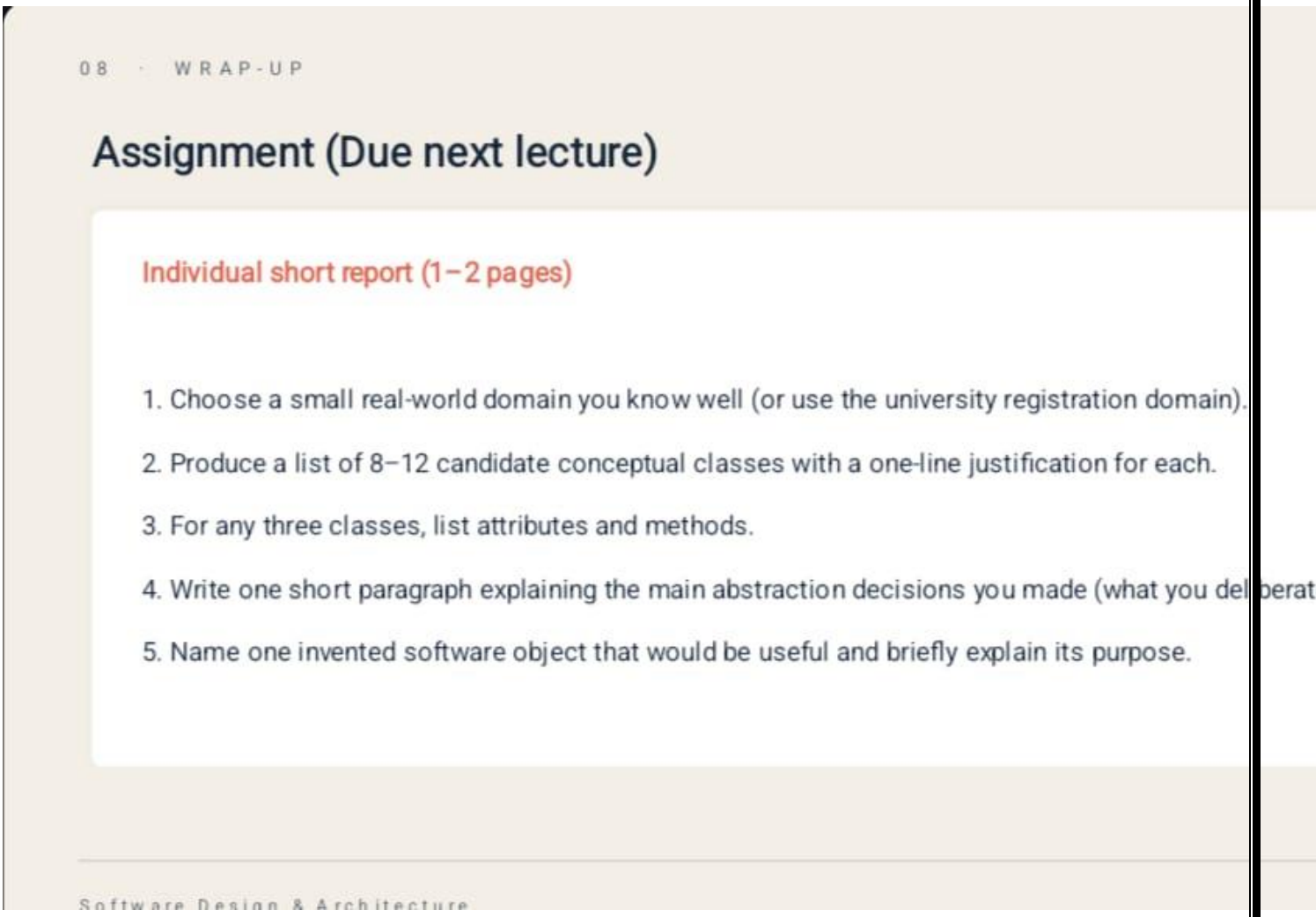
The University Of Azad Jammu and Kashmir, Muzaffarabad

Department of Software Engineering



Student Name	**Muhammad Zain Mughal**
Roll Number	**2025-SE-03**
Program	**BS Software Engineering**
Course Title	**Software Design & Architecture**
Lab Title	**From Real World Concepts to Software Object**
Assignment No	**01**
Course Instructor	**Engr. GR Butt**
Lab No.	**02**
Semester	**3rd**
Session	**2025–29**
Date	**19-Oct-2026**

Assignment



I have solved it step by step in detail; as follow

Task 1 — Choose a Domain

I chose Hospital Management System / Hospital Appointment.

Task 2 — List 10 Classes

Here are 10 conceptual classes that I listed below for hospital system:

1. Doctor
2. Patient
3. Medical Record
4. Admission
5. Appointment
6. Payment
7. Bill
8. Prescription
9. Room

10. Department

Task 3 — Write Attributes and Methods for 3 Important Classes

1. Patient

Attributes

- name: string
- patient ID: int
- contact: int
- blood group: char

Methods

- book_appointment
- cancel_appointment
- view_appointment
- check_Medical Report

2. Doctor

Attributes

- name: string
- doctor ID: int
- specialization: string
- contact Number: int

Methods

- view Appointments
- update Medical Report
- write prescription

3. Appointment

Attributes

- appointment ID: int
- status: string
- patient details: string
- doctor Name: string

Methods

- schedule appointments
- cancel Appointment
- check Appointment

Task 4 — Abstraction Decision (3–4 Sentences)

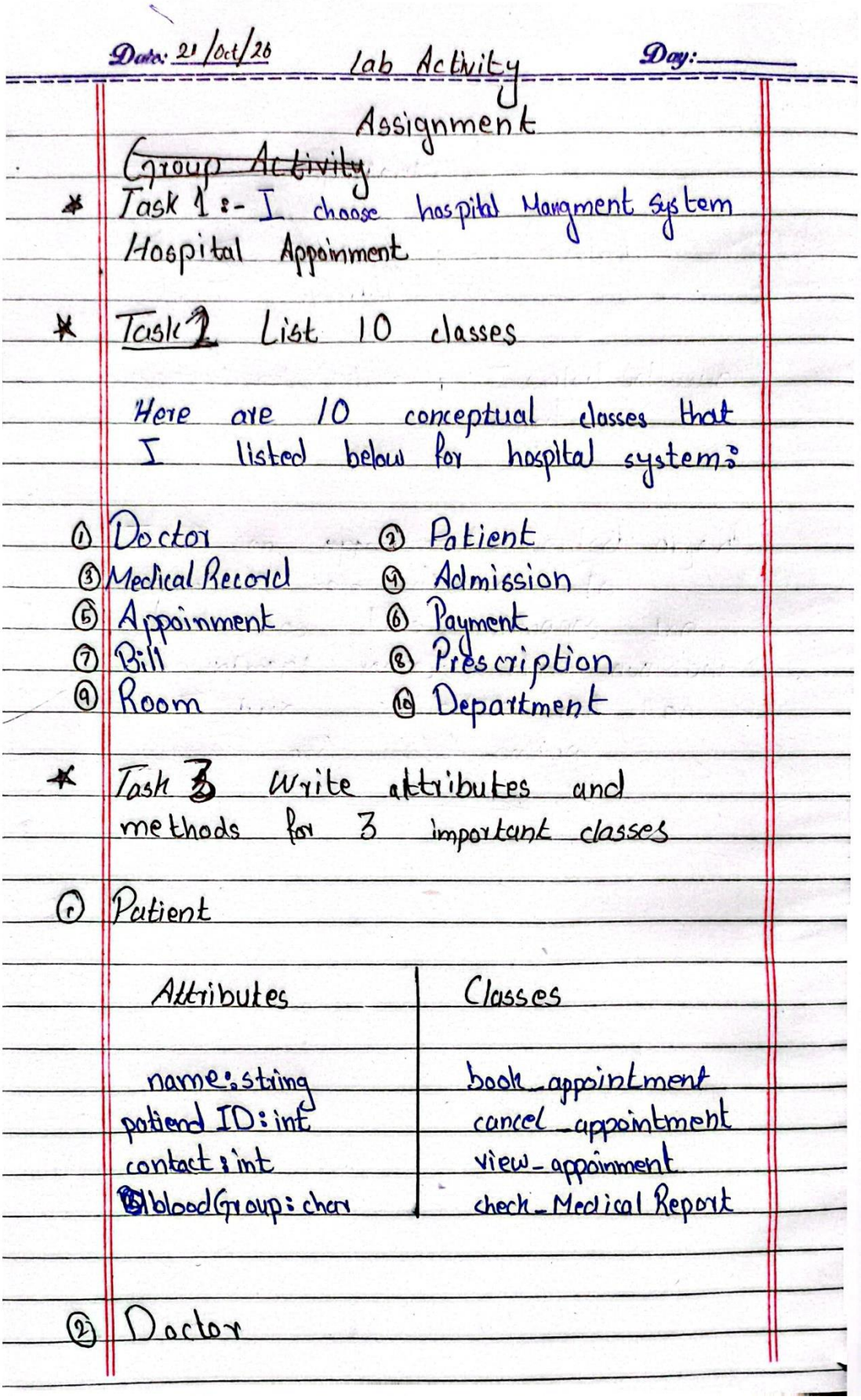
In this, I have only added the important details like patient ID, doctor ID, appointment status, appointment time, etc. because these details are required to run a hospital management system. I ignore the irrelevant information such as skin colour, hobbies, etc. because these are not important.

Task 5 — Identify One Invented Software Object

Appointment Controller → Invented Software Object

It is an invented object because it helps software to work. This manages the appointment request, checks availability and time schedule, and then confirms and controls booking.

I have attached the screenshots of hand written work also



Date: 21/Oct/26 Lab Activity Day:

Assignment

~~Group Activity~~

* Task 1 :- I choose hospital Mangment system Hospital Appoinment

* Task 2 List 10 classes

Here are 10 conceptual classes that I listed below for hospital system:

1. Doctor
2. Patient
3. Medical Record
4. Admission
5. Appoinment
6. Payment
7. Bill
8. Prescription
9. Room
10. Department

* Task 3 Write attributes and methods for 3 important classes

1. Patient

Attributes	Classes
name: string	book_appointment
patient ID: int	cancel_appointment
contact: int	view_appoinment
bloodGroup: char	check_Medical Report

2. Doctor

Date: _____ Day: _____

Attributes	Classes
name: string doctor ID: int Specialization: string Contact Number: int	view Appoinments update Medical Report write prescription

③ Appoinment

Attributes	Methods
appoinment ID: int status: string patient details: string doctor Name: string	schledule appoinments cancel Appoinment check Appoinment

* Task 3: 3 to 4 senteces on abstraction decision.

In this i have only add the important details like patient id, doctor id, appoinment status, appoinment time, etc because these details are required to run a hospital managment system. I ignore the irrelevant information such as skin color, hobbies etc because these

Date: _____ Day: _____

are not important.

* Task 5 Identify one "invented" software object.

Appointment Controller ⇒ Invented Software Object.

It is an ~~appointment~~ invented object because it help software to work as this manages the appointment request, check availability and time schledule and then confirm and control booking



This demonstrated how a real-world hospital domain can be represented through conceptual classes.
